SHUBHAJIT NANDI

Senior Backend Engineer | Distributed Applications & Messaging

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SUMMARY

Senior backend engineer with 11+ years across messaging, telecom protocols, payment services, and SaaS platforms. Built an SMSC handling 3,000 TPS and Kafka data pipelines that reduced processing latency by approximately 50%. Current work includes Spring Boot architecture, cloud delivery, and leadership of a five-engineer team.

TECHNICAL SKILLS

Backend & Architecture: Java, Python, C, Spring Boot, FastAPI, microservices, event-driven architecture, CQRS, Axon Framework

Messaging & Pipelines: Apache Kafka, Kafka Connect, NATS JetStream, AWS SQS

Data & Availability: PostgreSQL, MySQL, MongoDB replica sets, Redis

Cloud & Operations: AWS, Terraform, Docker, Kubernetes, GitHub Actions, Grafana, Grafana k6

Telecom & Voice: SMPP, SS7, SCTP, SIP, Asterisk, LiveKit, Twilio

PROFESSIONAL EXPERIENCE

Aspire Tech Services and Solutions Corp. (US-based client)

Senior Software Engineer | Jun 2025 - Present | Remote, Dhaka

- Lead a five-engineer team delivering ASAT-V2, a security-awareness SaaS platform, owning backend architecture, Spring Boot services, and production delivery.
- Built Spring Boot services for authentication, billing, phishing simulations, notifications, and reporting; integrated Stripe and PayPal with coupon, VAT, and discount handling.
- Configured a three-member MongoDB replica set with automatic failover for the SaaS platform.
- Built email, SMS, and voice phishing simulations, integrating Twilio and ElevenLabs and capturing participant responses for security-awareness risk reporting.
- Own the production environment and DevOps delivery for ASAT-V2. Set up development, test, and staging servers, configure deployment environments, and build GitHub Actions CI/CD pipelines for service deployment and versioned rollback.
- Used Grafana k6 for load testing and performance-metrics analysis of backend services.
- Reduced JavaScript asset loading time by approximately 50% by moving asset delivery to Amazon S3 and CloudFront.

Verbex

Senior Software Engineer | Aug 2023 - Jun 2025 | Dhaka, Bangladesh

- Built a Kafka and Kafka Connect data-warehousing pipeline, reducing processing latency by approximately 50% for downstream analytics.
- Developed a custom Kafka Connect transformation plugin to reshape complex source data for BI dashboards.
- Built seven Spring Boot services for a conversational voice AI platform using Java 21, Axon Framework, and CQRS, covering agent management, call orchestration, and transcription analysis.
- Integrated LiveKit SIP trunking to support inbound and outbound calls for the conversational AI platform.

Dotlines Bangladesh

Senior Software Engineer | Nov 2020 - Aug 2023 | Dhaka, Bangladesh

- Built a C-based SMSC using SMPP, SS7, and SCTP, handling 3,000 TPS and reducing per-message routing costs by 90% compared with third-party providers.
- Developed an asynchronous outbound-call gateway using Python, FastAPI, asyncio, and NATS JetStream to increase parallel call-processing capacity.
- Developed Java payment-gateway services deployed on Grameenphone and Banglalink platforms, integrating multiple third-party payment providers.

SSD-TECH

Software Engineer | Feb 2015 - Nov 2020 | Dhaka, Bangladesh

- Developed SMS delivery systems in C and PHP with MySQL and Redis, running on Banglalink's production platform at 2,000 TPS.
- Built reusable SMPP and SIP libraries in C for multiple telecom projects; implemented call-patching and conference-call features using Asterisk.

EDUCATION

Bachelor's in Computer Science and Engineering | 2010 - 2014

Chittagong University of Engineering and Technology (CUET), Chittagong, Bangladesh

CERTIFICATION & LANGUAGES

Splunk Certified Cybersecurity Defense Analyst

Bengali: Native | English: Fluent